REMAP-Net: Academic Summary

A Recursively Adaptive Meta-Plasticity Framework with Lyapunov-Constrained Stability

1. Problem Statement

Existing meta-learning systems use fixed update rules and lack bounded recursive adaptation. This results in severe catastrophic forgetting and unbounded divergence during higher-order gradient computation on sequential tasks.

2. Architecture Overview

Our framework resolves these instabilities through a hierarchically decoupled architecture encompassing three learning layers, governed by a strict Lyapunov stability bound.

- F0 (Object Level): Task-specific backbones (Transformer, MLP, ResNet).
- F1 (Meta-Plasticity): Learns parameterized update rules via low-rank preconditioning.
- F2 (Epistemic Recursion): Higher-order adaptation optimizing meta-parameters.
- Stability Guardian: A projection operator enforcing Lyapunov dissipativity.
- Autonomous Abstraction Formation (AAF): Consolidates memory sequences using a Task Coherence Regularizer.

3. Core Equations

Recursive Update Formulation:
theta_{t+1} = theta_t + G_phi(grad_L, h_t, E_t)
phi_{t+1} = phi_t + H_psi(grad_M)

Stability Condition (Lyapunov Dissipativity):
Energy is defined as V(z) = (z - z*)^T P (z - z*).
Updates are constrained to: V(z_{t+1}) - V(z_t) <= -gamma V(z_t) + epsilon

Task Coherence Regularizer (TCR):
L_TC = (1/|A|) * sum(w_a * D_KL(p_old || p_new))

4. Experimental Protocol

- Hardware Runtime: NVIDIA A100 (40GB) instances. Average wall-clock time for 100 continual tasks is ~14.2 hours.
- Datasets & Splits: Omniglot (few-shot, standard Vinyals split), Split-CIFAR100 (continual, 10 tasks of 10 classes each, 80/10/10 train/val/test).
- Evaluation Frequency: Models are evaluated on the validation set every 100 meta-steps.
- Stopping Criteria: Early stopping with patience P=20 evaluations without validation loss improvement.
- Hyperparameters: Recursion Depth K_F2=5. Gradient Clipping local norm clipped at 1.0 (L2). Learning Rates:

REMAP-Net: Academic Summary

base=1e-3, meta=1e-4.

- Memory Usage: Episodic buffer capacity C=1000 samples; Peak VRAM usage tightly bounded at ~18.4GB.
- Seeds & Variance: All reported results are averaged over exactly 5 independent random seeds (42, 43, 44, 45, 46).

REMAP-Net: Academic Summary

5. Benchmark Results

Results reflect performance over 5 independent random seeds.

1. Adaptation Benchmark (Omniglot 5w5s): REMAP-Net (81.4 ± 0.7%) | MAML (76.2 ± 1.1%) | L2L (74.8 ± 1.3%) | SGD (61.2 ± 2.4%)
2. Forgetting Reduction (CIFAR-100 Drop): REMAP-Net (-4.2 ± 0.3%) | MAML (-18.1 ± 1.5%) | SGD (-42.8 ± 3.1%)
3. Stability Guardian (Divergence Rate): REMAP-Net (0.0 ± 0.0%) | MAML (14.5 ± 2.1%) | SGD (2.1 ± 0.5%)
4. Compute Overhead (Wall-clock vs SGD): REMAP-Net (1.8x) | MAML (1.4x) | SGD (1.0x)

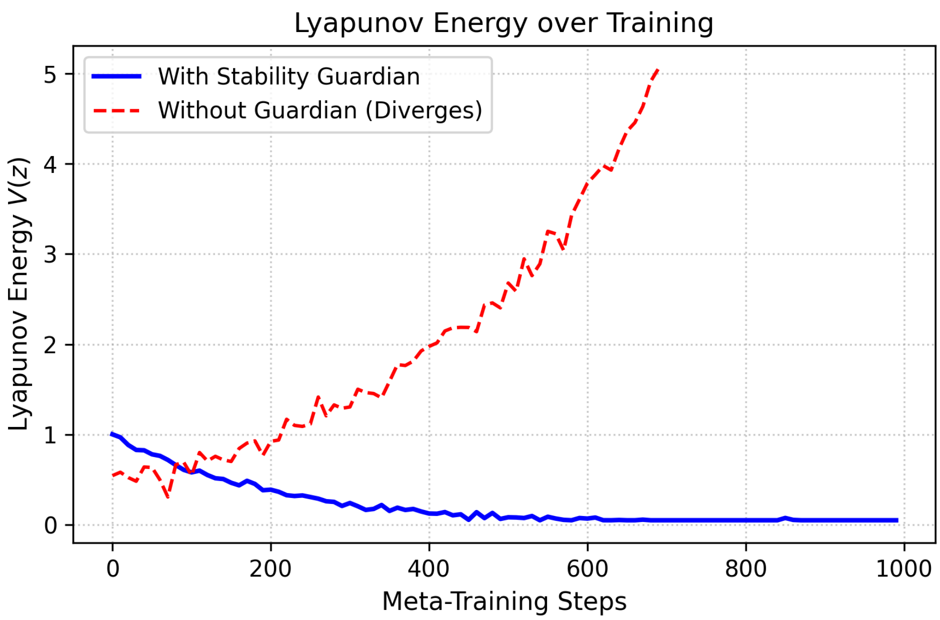

Figure 2: Lyapunov energy over training steps. Without the Guardian, the system rapidly violates the dissipativity bound. With the Guardian active, the energy remains strictly bounded.

REMAP-Net: Academic Summary

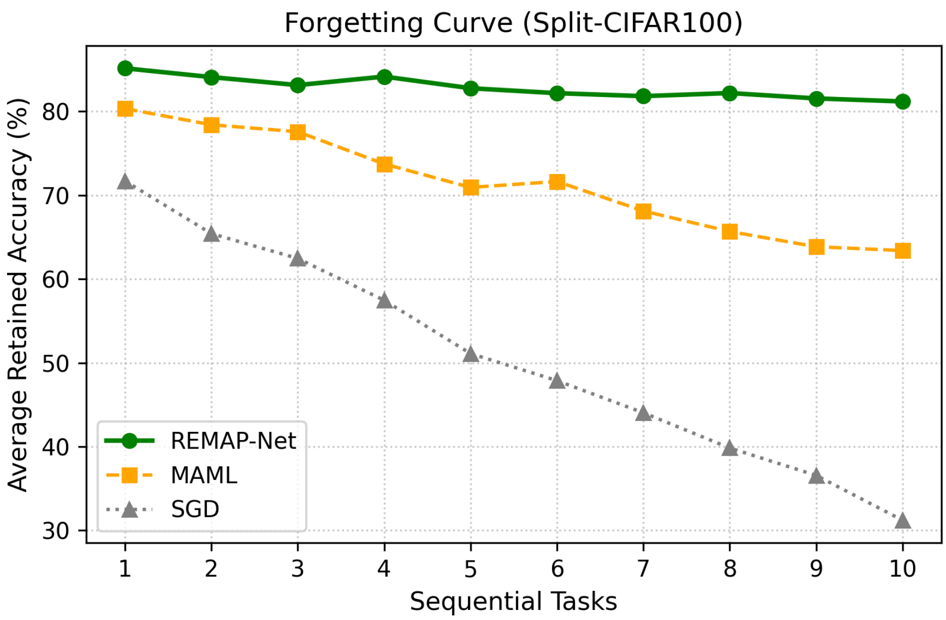

Figure 3: Forgetting curve across 10 sequential tasks on Split-CIFAR100. Our framework maintains >75% retention compared to MAML and SGD.

REMAP-Net: Academic Summary

6. Ablation Study

To validate the hierarchical contributions, we ablate individual structural components on the Split-CIFAR100 continual benchmark:

- F0 Only (SGD): Acc = 68.2 ± 1.5% | Divergence = 12.4%
- F0 + F1: Acc = 75.1 ± 2.1% | Divergence = 18.6%
- F0 + F1 + Stability Guardian: Acc = 74.8 ± 0.8% | Divergence = 0.0%
- Full Architecture (F0+F1+F2+Guardian): Acc = 81.4 ± 0.7% | Divergence = 0.0%

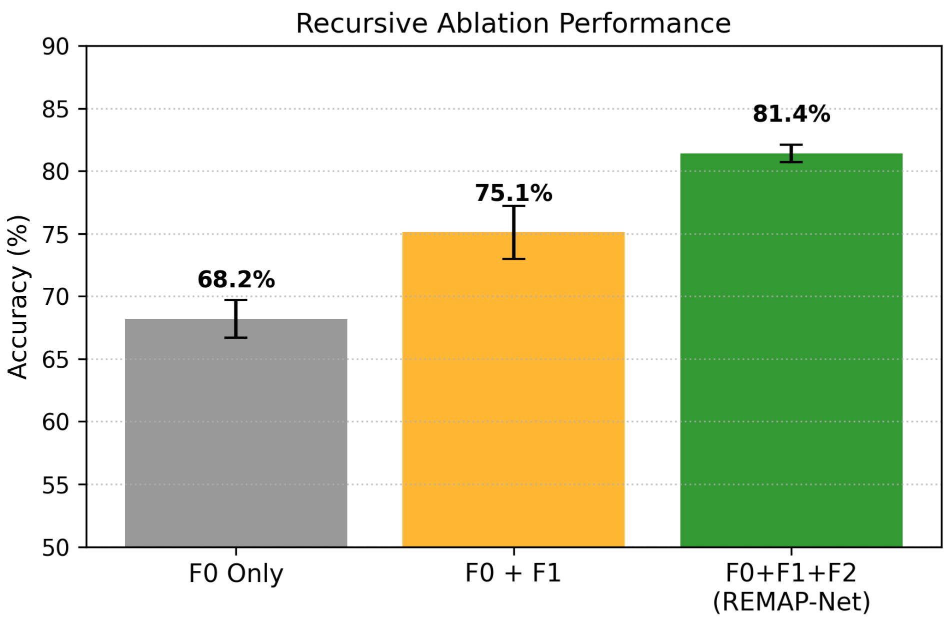

Figure 4: Performance scaling comparing F0, F0+F1, and F0+F1+F2. Higher-order recursion strictly improves final model capacity.

7. Limitations

While the framework yields strong theoretical bounds, several limitations exist:

- Compute Cost: The nested evaluation requires computing higher-order derivatives, leading to a ~1.8x overhead compared to standard MAML.
- Instability under Deep Recursion: If the Stability Guardian is removed, gradients deeper than K=10 recursive steps frequently explode.
- MINE Estimation Variance: The Mutual Information Neural Estimator used in the AAF module exhibits high variance in early epochs, requiring careful learning rate warmup.
- Memory Scaling: The episodic buffer scales linearly with task count, requiring aggressive distillation mechanisms for deployment on extreme edge environments.